

**Part—2**

The first article of this series (in the February 2012 issue of *LFY*) gave an introduction to Kinect. In this second part, I cover the installation of *libfreenect*. Since these articles are also addressed to newbies, do pardon the inclusion of some of the basics.

For programming with Kinect, we need OpenCV (Open Source Computer Vision), a library for real-time computer vision; so let me show you how to install this on Ubuntu 10.04, along with *libfreenect*.

To install OpenCV dependencies, use the following commands:

```
sudo apt-get install build-essential libgtk2.0-dev libavcodec-dev
libavformat-dev libjpeg62-dev libtiff4-dev cmake libswscale-dev
libjasper-dev
cd ~
```

Download OpenCV-2.1.0 from <http://bit.ly/cZLUzS> and extract it. Change to the extracted folder. After the *cmake* step, check whether the *ffmpeg* value is 1. If not, run it again.

```
cd OpenCV-2.1.0/
cmake .
make
sudo make install
sudo gedit /etc/ld.so.conf.d/opencv.conf
```

At the end of the file, add *'/usr/local/lib'*, then save it, and run the following:

```
sudo ldconfig
sudo gedit /etc/bash.bashrc
```

Add the following two lines at the end of the file (it should not be empty):

```
PKG_CONFIG_PATH=$PKG_CONFIG_PATH:/usr/local/lib/pkgconfig
export PKG_CONFIG_PATH
```

Now OpenCV-2.1.0 has been installed. Open another terminal to check some of the samples:

```
cd ~
cd OpenCV-2.1.0/samples/c
chmod +x build_all.sh
./build_all.sh
```

To check any sample, go to that folder, and run the sample. For example, in the *c* folder, we have *contours2*, so to run it:

```
cd c
./contours2
```

Libfreenect

To install *libfreenect*, we need to install some prerequisites for building it:

```
sudo apt-get install git-core cmake libglut3-dev pkg-config build-
essential libxmu-dev libxi-dev libusb-1.0-0-dev
```

Next, clone its source from *Github*, change into the *'libfreenect'* directory, create a *'build'* sub-directory and *cd* to it, then run *cmake* to generate native *Makefiles* and workspaces that can be used in the compiler environment of your choice:

```
git clone git://github.com/OpenKinect/libfreenect.git
cd libfreenect
```